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Centre Number

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Student Number

NSW Education Standards Authority

IMPORTANT:

This resource was developed to support a previous version of the syllabus.
It may contain content that differs from the current syllabus.

2023 HIGHER SCHOOL CERTIFICATE EXAMINATION

Information Processes and Technology

**General
Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Write your Centre Number and Student Number at the top of this page

**Total marks:
100**

Section I – 20 marks (pages 2–7)

- Attempt Questions 1–20
- Allow about 40 minutes for this section

Section II – 40 marks (pages 8–11)

- Attempt Questions 21–24
- Allow about 1 hour and 10 minutes for this section

Section III – 40 marks (pages 12–16)

- Attempt TWO questions from Questions 25–28
- Allow about 1 hour and 10 minutes for this section

Section I

20 marks

Attempt Questions 1–20

Allow about 40 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

- 1** An error was found in an app after implementation.

In which stage of system development was this error found?

- A. Planning
- B. Designing
- C. Understanding the problem
- D. Testing, evaluating and maintaining

- 2** A project manager and a client discuss issues and compromise to reach a solution.

Which communication technique are they demonstrating?

- A. Interview
- B. Surveying
- C. Negotiation
- D. Team building

- 3** Which of the following project management tools outlines the process for how information should be distributed, and who it should be delivered to?

- A. Gantt charts
- B. Journals and diaries
- C. Funding management plan
- D. Communication management plan

- 4 What is a benefit of using a wireless network in a small business?
- A. Better data security than a wired network
 - B. Easy sharing of data between multiple devices
 - C. Unlimited transmission area with a single router
 - D. Higher transmission speed than a wired network
- 5 Which of the following should be considered when determining the operational feasibility of a proposed system?
- A. Cost of developing the system
 - B. Time taken to implement the system
 - C. Users having the skills needed to use the system
 - D. Availability of information technology for the system
- 6 Which of the following would create the most stable network with the fastest data transmission?
- A. Use a wireless access point to connect devices
 - B. Connect devices wirelessly using a mobile phone
 - C. Use coaxial cables to connect devices in sequence
 - D. Connect devices to a switch using optic fibre cables
- 7 Which statement describes how a search robot works on a search engine?
- A. Keeps a list of URLs for web pages containing keywords
 - B. Sends appropriate URLs back to the indexing database
 - C. Provides a directory structure for the user
 - D. Conducts keyword searches

- 8** Documentation for the user is created at different stages of system development.

Which statement best describes the purpose of user documentation created during the implementation stage?

- A. Recording the development process of the system
- B. Listing the user requirements of the developed system
- C. Providing help and instructions for the use of the system
- D. Gaining feedback from users and participants of the system

- 9** A bank is going to use a computer system to automatically respond to customer enquiries. During development, a small range of data is used to test the system.

Which type of testing is this?

- A. Hardware
- B. Live
- C. Simulated
- D. Volume

- 10** What is a disadvantage of BOTH satellite and infrared transmission?

- A. Implementation cost is low
- B. Direct line of sight is required
- C. Direct wired connection is required
- D. Transmission speed over long distance is high

- 11** Which of the following statements best describes a blog?

- A. A website that allows multiple users to collaborate and create information
- B. A chronological list of posts on a website typically written by a single author about a subject
- C. A series of audio or video episodes that subscribers download or stream to their devices
- D. A standardised format that allows users to subscribe to view content updates from a creator

- 12 Which of the following best describes the Ethernet protocol?
- A. Infrared or microwave signals are used to transmit data.
 - B. Data is transmitted by sending light signals through wires.
 - C. Data is transmitted between a sequence of nodes on a network.
 - D. A medium is shared where all devices have access to transmit on a network.
- 13 How do HTTP and SSL protocols work together to create a secure connection?
- A. Encode and encrypt data between nodes
 - B. Use twisted pair and coaxial cables for transmission
 - C. Authenticate and synchronise data during network logon
 - D. Define an address and establish a route between the recipients
- 14 A local council wants to find out from its residents the issues that are most important to them. An online survey is to be designed to collect data. Each resident will be presented with a list of issues and they can select three issues from the list relevant to them.

Which of the following screen elements should be used to capture the response?

- A.
- | Issue list | Select three issues |
|------------|-----------------------|
| Issue 1 | ☐ |
| Issue 2 | ☐ |
| Issue 3 | ☐ |
| Issue 4 | ☐ |
- B.
- | Issue list | Select three issues |
|------------|--------------------------|
| Issue 1 | ☐ |
| Issue 2 | ☐ |
| Issue 3 | ☐ |
| Issue 4 | ☐ |
- C.
- | Issue list | Select three issues |
|------------|----------------------|
| Issue 1 | <input type="text"/> |
| Issue 2 | <input type="text"/> |
| Issue 3 | <input type="text"/> |
| Issue 4 | <input type="text"/> |
- D.
- | Issue list | Select three issues |
|------------|----------------------|
| Issue 1 | <input type="text"/> |
| Issue 2 | <input type="text"/> |
| Issue 3 | <input type="text"/> |
| Issue 4 | <input type="text"/> |

KEY

Radio Button ☐ Check box ☐ Text box Drop down

- 15 What is the purpose of synchronising the exchange of data when transmitting between a source and a destination?
- A. Managing the transmission of data between the devices
 - B. Reducing the amount of data transmitted to increase speed
 - C. Ensuring that data packets are delivered using the fastest route
 - D. Preventing data from being lost or corrupted during transmission
- 16 Which row of the table identifies approaches that could be used to identify problems within an existing system?
- A. Interview participants Analyse the system
 - B. Survey users Develop a conversion method
 - C. Conduct a feasibility study Analyse the system
 - D. Create a requirements report Develop a conversion method
- 17 The following is a decision table outlining different insurance product options for customers.

<i>Type of cover (conditions)</i>	<i>Rules</i>			
Injury to drivers and passengers	Y	N	N	N
Injury to others	Y	N	N	N
Fire	N	Y	Y	Y
Theft	N	Y	Y	Y
Choice of repairer	N	N	N	Y
Rental car	N	N	N	Y
<i>Insurance product (actions)</i>				
Basic	Y	N	N	N
Silver	N	Y	N	N
Gold	N	N	Y	N
Platinum	N	N	N	Y

Which of the following is correct based on the information provided?

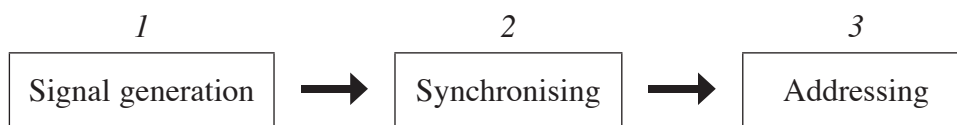
- A. Basic insurance covers choice of repairer and injury to others.
- B. Silver insurance covers fire and choice of repairer.
- C. Gold insurance covers fire and theft.
- D. Platinum insurance covers all types of injuries and a rental car.

- 18 A librarian wants to use SQL to generate a list of people who borrowed books in January 2023, including the book title and author.

Which combination of SELECT and WHERE statements will achieve this?

- A. SELECT Name, Title, Author
WHERE Date_Borrowed >= 1/1/2023 AND Date_Borrowed <= 31/1/2023
- B. SELECT Name, Date_Borrowed, Title, Author
WHERE Date_Borrowed <= 1/1/2023 AND Date_Borrowed >= 31/1/2023
- C. SELECT Name, Title, Author
WHERE Date_Borrowed < 1/1/2023 AND Date_Borrowed > 31/1/2023
- D. SELECT Name, Date_Borrowed, Title, Author
WHERE Date_Borrowed > 1/1/2023 AND Date_Borrowed < 31/1/2023

- 19 The following diagram represents a portion of a communication system.



Which row of the table correctly identifies appropriate hardware for the three stages?

	1	2	3
A.	Network interface card	Gateway	Bridge
B.	Network interface card	Modem	Wireless access point
C.	Hub	Gateway	Bridge
D.	Hub	Modem	Wireless access point

- 20 Which row of the table identifies a difference between a *switch* and a *router* in a computer network?

	<i>Switch</i>	<i>Router</i>
A.	Connects multiple nodes within a single network	Connects multiple networks of the same protocols
B.	Operates at the communication control and addressing level	Operates at the transmission level
C.	Provides network security	Provides network stability
D.	Allows for fast transmission of data between different networks	Allows for fast transmission of data between nodes

Section II

40 marks

Attempt Questions 21–24

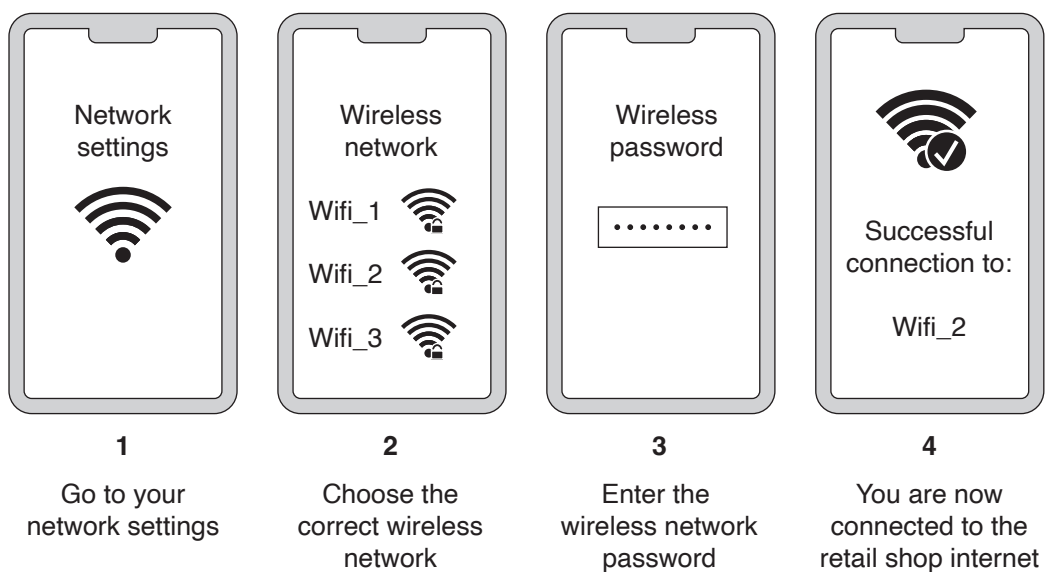
Allow about 1 hour and 10 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

If you include diagrams in your answer, ensure that they are clearly labelled.

Question 21 (9 marks) Use the Question 21 writing booklet.

- (a) Describe the role of a network administrator. **3**
- (b) Compare the use of thin clients and fat clients on client-server networks. **3**
- (c) A retail shop provides free access to their wireless internet connection. The steps that visitors of the shop use to connect to the wireless network are shown. **3**



Describe *handshaking* in this scenario.

Question 22 (10 marks) Use the Question 22 writing booklet.

- (a) An internet service provider has a limited budget allocated to design a new system. The proposed system is intended to be released in one month's time. The system will run an automated query every two weeks to check its customers' data usage. Customers who exceed 100% of their data usage will be charged for excess data and experience slow connection. Customers will be able to log into their online account, check their data usage and pay their monthly bill.
- (i) Describe the scheduling and economic risks that should be considered in the feasibility study of this new project. **3**
 - (ii) Explain how a requirements report could help the internet service provider to develop the system. **4**
- (b) Compare traditional and iterative system development approaches. **3**

Please turn over

Question 23 (10 marks) Use the Question 23 writing booklet.

A messaging system allows users to exchange text messages online. User details and messages are stored in a flat-file database. A record from the flat-file database is shown.

SenderPhone	0312 621 981
ReceiverPhone	0815 683 665
Message	Meeting is at 10 am
MessageTime	31/01/2023 16:17
MessageRead	☒

Both the senders and receivers are users of the messaging system.

- (a) Copy the following table into your writing booklet.

3

In your table, recommend an appropriate field size and data type for SenderPhone and MessageRead. Justify each choice of data type.

<i>Field</i>	<i>Field size</i>	<i>Data type</i>	<i>Justification for choosing the data type</i>
SenderPhone			
MessageRead			

- (b) The flat-file database is to be converted into a relational database. In addition to the phone number, more information such as name and email address will be stored for each user.

3

Draw a schematic diagram to show how the database can be normalised. Clearly label the names of the tables, primary and foreign keys and any relationships.

- (c) The messaging system will be further enhanced to allow users to send images and attach files.

4

Describe ethical and technical issues that may result from this further enhancement.

Question 24 (11 marks) Use the Question 24 writing booklet.

- (a) The following data packets were sent and received.

3

<i>Sent by customer</i>	<i>Received by system</i>
11001111 00111110	00111110 11001111

Explain why cyclic redundancy check (CRC) will find an error in the received data but check sum will not.

- (b) A large technology company is developing an online booking system. This system will allow customers to logon and make bookings for the repair of their electronic devices which were manufactured by the company.

Customers open the company's web page from a computer or mobile phone and fill in the booking form with their personal details, payment information and service requirements. These include the time and date of booking, hardware model, software version, service required and other relevant details.

The system will send a confirmation email to the customer with their booking details. Data from the system is stored in the company's data warehouse for future data mining.

- (i) Draw a data flow diagram that represents the technology company's online booking system.

3

- (ii) Explain how the data stored in the data warehouse can be used by both the technology company and their customers.

5

Section III

40 marks

Attempt TWO questions from Questions 25–28

Allow about 1 hour and 10 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

If you include diagrams in your answer, ensure that they are clearly labelled.

Question 25 — Transaction Processing Systems (20 marks)

Use a SEPARATE writing booklet.

- (a) Describe why both secure on-site and off-site storage are important when backing up transaction data. Include an example in your answer. **3**

- (b) A website allows customers to purchase original car parts and accessories. The website connects to all major car manufacturers by displaying their available stock, and manages customer orders. The website allows customers to register an account, order and pay for their purchases via the online store, and allows a customer to pick up purchases in-store or select a different delivery method.
 - (i) Describe the information technology required for this transaction processing system. **3**
 - (ii) Explain the importance of data quality and data integrity for this transaction processing system. **4**

- (c) Describe an example where an online analytical processing (OLAP) and an online transaction processing (OLTP) system would be used together. **4**

- (d) A mobile food truck owner implements a contactless payment system. This system allows them to connect a hardware device to their personal mobile phone for accepting payments of under \$100.00 from credit and debit cards, without the need for a customer passcode. The hardware device reads the information from the card using RFID technology and instantly transfers the money to the owner's bank account. **6**

Explain how the issue of data security can be managed in this transaction processing system. In your response, include reference to the information processes of *collecting*, and *storing* and *retrieving*.

Question 26 — Decision Support Systems (20 marks)

Use a SEPARATE writing booklet.

- (a) Describe why structured decision making does not require a decision support system. Include an example in your answer. **3**

- (b) Doctors diagnose patients by observing a patient's symptoms and asking them questions. **3**

Describe the role of the inference engine in an expert system that helps doctors diagnose patients.

- (c) The following spreadsheet shows part of a payroll system. **4**

	A	B	C	D	E	F
1	Australian Tax Rates					
2	Income	Base tax	Tax rate		Employer superannuation contribution rate	
3	\$18 201.00	\$0.00	\$0.190		10.50%	
4	\$45 001.00	\$5 092.00	\$0.325			
5	\$120 001.00	\$29 467.00	\$0.370			
6	\$180 001.00	\$51 667.00	\$0.450			
7						
8	Employee	Annual income	Base tax	Extra tax	Total amount of tax payable	Superannuation contribution
9	Alex Smith	\$125 000.00	\$29 467.00	\$1 849.63	\$31 316.63	\$13 125.00
10	Charlie Antwe	\$95 200.00		\$16 314.68	\$21 406.68	\$9 996.00
11	Jesse Singh	\$33 500.00	\$0.00	\$2 906.81		

The base tax for an employee is determined from on their annual income. For example, the base tax for an employee earning between \$120 001 and \$180 000 is \$29 467. The total amount of tax payable for each employee is calculated by adding the base tax and the extra tax. The employer superannuation is calculated by multiplying the superannuation contribution rate and the annual income.

Design formulae for cells C10, E11 and F11 so they can be copied into multiple cells.

Question 26 continues on page 14

Question 26 (continued)

- (d) A chatbot, created using neural network technology, is designed to conduct conversations with humans. The chatbot was initially ‘taught’ from a wide range of academic articles. The chatbot improves responses by seeking feedback from its users. **4**

Explain why the use of a neural network in the chatbot’s learning affects its ability to converse with humans.

- (e) A new geographical information system (GIS) is being developed to represent different walking trails used by bushwalkers. It provides information on where to find and book picnic facilities as well as providing weather and other environmental conditions. Information is updated regularly on whether walking trails are accessible or facilities are available to be booked. **6**

Explain how the use of a geographical information system in this scenario allows for the analysis of unstructured situations. In your response, include reference to the information processes of *analysing* and *displaying*.

End of Question 26

Question 27 — Automated Manufacturing Systems (20 marks)

Use a SEPARATE writing booklet.

- (a) Describe the process of converting analog data to digital data in an automated manufacturing system. Include an example of a device that converts analog data to digital data. **3**
- (b) Using an example, describe how the integration of sensors into manufacturing machinery can automate processing. **3**
- (c) Compare the use of a solenoid and a relay as an actuator. **4**
- (d) An automated system is designed to maximise solar energy collection by keeping solar panels directly facing the sun throughout the day. The system consists of sensors, a control unit and actuators that perform the tasks of detecting and tracking the location of the sun and adjusting the angles of solar panels to face the sun. After sunset the solar panels are returned to the morning starting position.
- (i) Outline the functions of the sensors, controller and actuators in the automated solar energy system. Use a block diagram to support your answer. **4**
- (ii) Explain how the issues of *precision* and *acceptable tolerance range* can be managed within the system. Include reference to the following information processes to support your response: **6**
- analysing
 - displaying.

Question 28 — Multimedia Systems (20 marks)

Use a SEPARATE writing booklet.

- (a) Describe the role of a storyboard when designing a multimedia product. Include an example in your answer. **3**

- (b) A software application is developed which allows people to experience virtual camping and hiking. **3**

Justify the use of .Wav and .MPG file types to store audio and video data in this application.

- (c) A business is designing an advertisement for its product. It is considering a printed version in a magazine or a multimedia version on an electronic sign that can be displayed on the side wall of a building. **4**

Describe advantages and disadvantages of presenting the advertisement in printed and multimedia versions.

- (d) A furniture company has created an application enabling customers to design and visualise their own rooms before purchasing items. Customers upload several photos of their existing room and the application adds the images to an interactive video display. Customers can also: **4**

- add furniture, paint colours and carpet from the company's catalogue
- export designs into an interactive video.

Explain how video data and frame rates can affect the processing required by this application.

- (e) Some contemporary websites enable their users to create an avatar (a 3D image of themselves) which can interact with other users in high quality, virtual environments. **6**

Users need to upload images at different angles of themselves and use web-based authoring software to create the avatar.

Explain how authoring software can be used to organise and display the avatars in this system. Include reference to the effects of bit depth and the representation of colour data.

End of paper